Injury Prediction and Player Safety in the NFL

Technical Report

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Datasets Used:

Video Review (NFL Punt Analytics)

Injury Record (NFL 1st and Future)

Concussions in the NFL

Topic Description:

This project uses publicly available National Football League (NFL) data to improve the health and safety of NFL players. This project will develop insights into common injuries and determine what aspects of the game increase a player's risk for injury, with a particular focus on concussions.

Section 1: Executive Summary

This project is designed to focus on improving the health and safety of National Football League (NFL) players and to develop insights into what injuries are most prevalent and what aspects of the game may increase the risk of player injury or a specific injury type. The results will then be used to make data-driven recommendations to the NFL on what strategies will be most effective in minimizing the negative impact the sport has on its players' wellbeing during and after their careers.

This extensive analysis included three datasets. The first, "Video_Review" is from the "NFL Punt Analytics" dataset. This dataset is a collection of information that was created based on reviewable video evidence that outlines the events that resulted in a concussion during punt players in the NFL 2016-2017 season. The second dataset, "InjuryRecord" accounts for 105 lower-limbs injuries that occurred over two seasons during the regular NFL season and provides information on the surface the game occurred on and the number of days the player missed due to injury (or how severe it was). Lastly, the "Concussion" dataset, contains a list of concussion injuries that occurred in the NFL from the year 2012 to 2014.

Exploratory data analysis (EDA) and the use of different machine learning models determined that concussions have a clear relationship with features like number of snaps into the game when injury occurs, the week of the season the injury occurs, and a player's position. This is encouraging since these results can help the NFL consider rule changes like shorter games, shorter seasons, or even offer additional protection to specific positions if the rate of injury is higher. In addition, the analysis also determined that tackling or being blocked is a major source of concussions on punt plays and 'friendly fire' had a moderate correlation with increased risk of concussion on these plays as well. These factors may be able to be addressed with new coaching

techniques or personnel changes; for example, maybe the NFL investigates moving to 10 players instead of 11 on punts.

Section 2: Introduction & Problem Definitions

The National Football League (NFL) is known as one of the most dangerous sports when it comes to player health and safety, while playing the sport and after the conclusion of one's career. According to a research study published in 2023, it was discovered that American football players, particularly linemen, were found to be prone to early ageing "characterised by premature burden of chronic disease and reduced healthspan" (Grashow et al., 2023). The NFL has worked to address these issues by introducing rule changes, guardian caps to prevent concussions, redesigning equipment, and procedural changes like the introduction of the dynamic kickoff. These changes have produced success, as the NFL stated in the 2024 season they saw "concussions decreased 17% compared to last season and 12% compared to the 2021-2023 season" (National Football League [NFL], 2025). Although improvements have been made, independent research has shown that the data might not be as strong as the NFL originally reported. For example, Katherine O'Malley in her article "There's a way to deal with brain injuries in football. It isn't safety gear", it was reported that when two lab-based studies and one on-field study replicated the use of guardian caps the results showed no difference in players who wore them versus those who did not wear them. As a result, the threat to player safety remains prevalent in all facets of the game.

This project's objective is to model, quantify, and predict the factors in NFL games that contribute to player injury risk, specifically the risk for concussions and lower body injuries. In particular, the project will identify plays, positions, actions, and environmental factors associated with injury risk and quantify these factors to understand the severity of their impact. To do this, supervised and unsupervised machine learning models and statistical methods were used. The

results of this analysis will allow for the NFL to make more efficient and effective decisions when investing into injury prevention for their players and maximize the positive impact of these changes.

The project's analytical approach began with EDA of all three datasets to cleanse and gather more insight into what relationships are worth exploring further. From there, several predictive learning models will be utilized such as classification, regression, and clustering to identify linear, non-linear, and complex patterns in the datasets that are not obvious. Once completed, the models will be evaluated using different metrics. Success criteria in this project will involve the following metrics: RMSE, R^2, Pearson correlation coefficient, accuracy, and F1-scores. The relationships that reached the following performance measures were considered to indicate a strong relationship between the target feature and that specific variable.

- RMSE < 0.40
- R^2 (Coefficient of Determination) > 0.70
- Accuracy > 70%
- F1-scores > 0.70
- Pearson correlation coefficient > 0.70

All three datasets utilized in this project are small; the "Video_Review" dataset has 37 entries, the "InjuryRecord" dataset has 105, and the "Concussion" dataset has 390 entries after dropping the 'NAN' values from the dataset. This made analysis difficult as they all are prone to overfitting, especially the "Video_Review" dataset. In addition, the most recent dataset was published in 2019. Having a full dataset from more recent seasons, especially since the adaption of the dynamic kickoff, helmet re-design, and new penalties for contact to players head and

lower bodies, would provide the project with a more accurate insight and in turn recommendations.

Section 3: Data Overview

Data sources

Dataset one, "Video_Review", is a collection of information that was created based on reviewable video evidence that outlines the events that resulted in a concussion during punt players in the NFL 2016-2017 season. The target, 'Primary_Impact_Type', outlines if the concussion occurred from the impact of Helmet-to-Helmet, Helmet-to-Body, or Helmet-to-Ground. Dataset 2, "InjuryRecord", looks to determine the relationship between the playing surface and the injury and performance of NFL athletes. The "InjuryRecord" dataset accounts for 105 lower-limbs injuries that occurred over two seasons during the regular NFL season and provides information on the surface the game occurred on and the number of days the player missed due to injury (or how severe it was). The target in this case is 'surface' which lists the type of surface (synthetic or natural) the field was when the injury occurred. The last dataset, "Concussion", contains a list of concussion injuries that occurred in the NFL from the year 2012 to 2014. The data includes features such as 'Position', 'Pre-Season Injury?', 'Week of Injury', 'Weeks Injured', 'Games Missed', 'Reported Injury Type', 'Average Playtime Before Injury'. The target in this case will be "Reported Injury Type" which will be limited to just concussions.

Data collection method

All three datasets were accessed online through public downloadable files posted on Kaggle. The "Video_Review" and "InjuryRecord" datasets were posted by the NFL as a part of their Analytics Competition and the "Concussion" dataset was posted by a Kaggle user who accessed the data through the data.world website.

Data dictionary

Table 1: Data dictionary for all three datasets

Feature	Data Type	Description	Example Value
Primary_Impact_Type_encoded	Categorical	Impact source that caused the injury (i.e. concussion)	'Helmet-to-ground'
Friendly_Fire	Categorical	Primary impact results from contact between two players on the same team	'No'
Surface	Character string	Surface type at time of injury	'Natural'
BodyPart	Character string	Lower body part injured	'Knee'
Reported_Injury Type	Categorical	Type of injury that occurred ('concussion' or 'head')	'Concussion'
Total Snaps	Numerical	Number of snaps player played before before injury occurred	'10'

Data description

Table 2: Summary statistics, data types, and amount of data for all three datasets

Feature	Number of Entries	Mean	Median	Mode
Primary_Impact_Type_encoded	39	1.054	Helmet-to-body	'Helmet-to-body' and 'Helmet-to-helmet'
Friendly_Fire	39	0.514	No	No
Surface	105	0.543	Synthetic	Synthetic
BodyPart	105	1.724	Knee	Knee
Reported_Injury Type	325	0.185	Concussion	Concussion
Total Snaps	325	33.071	30.0	17.0

Data quality assessment

Table 3: Data quality assessment in all three datasets

Dataset	Feature	Missing Value Percentage	Handling Approach
'Video_Review'	'Primary_Impact_Type_Derived"	0%	None needed
'InjuryRecord'	'Surface'	0%	None needed
'Concussion'	'Reported Injury Type'	8.11%	Drop the rows with missing values

Ethical considerations

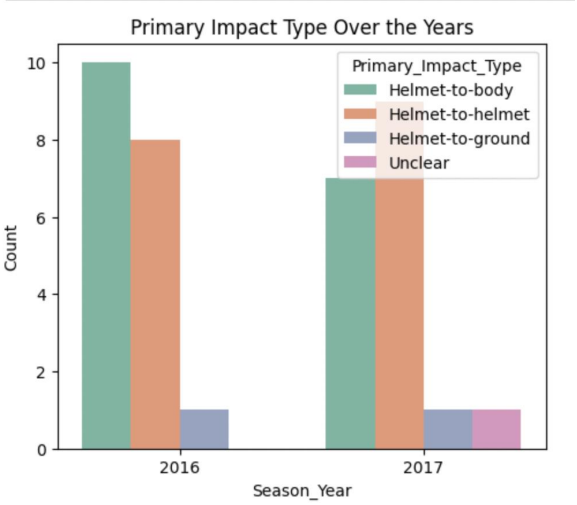
In the datasets "Video_Review" and "InjuryRecord", the player's information is unidentifiable. These datasets utilize ID numbers to characterize who the player is without providing their names, teams, or any other information. However, the "Concussion" dataset lists the player's name, team, position, and the game the injury occurred in. All these factors make it easy to determine who the player was and offer little privacy. In this analysis, all identifiable information was dropped except position which was found to be useful for reaching the project's objective. In addition, the datasets that were used ranged over a 5-year timespan; the "Concussion" dataset was from 2012-2014, the "Video_Review" dataset was from 2016 to 2017,

and the “InjuryRecord” was from what is suspected to be 2018. As a result, there is potential for selective inclusion or bias to either include seasons where injuries were frequent, or the datasets were published by the NFL to appease negative attention the sport faced regarding player health and safety. The data collected is from league or public datasets, so it is assumed player’s consent was obtained.

Section 4: Data Cleaning, Preprocessing, Exploratory Data Analysis (EDA)

During the preprocessing phase of the exploratory data analysis, it was determined that all three datasets had no duplicate rows and no duplicate columns. The analysis at first did produce information that made it seem as if there were duplicate columns, however after further analysis though, it was determined the analysis was looking at the specific data inside the table. The duplicate column inside the data is expected since the data was looking at specific features, such as year in the “Video_Review” dataset (i.e. 2016 or 2017) or the specific ‘surface’ type as seen in the “InjuryRecord” dataset which are not unique to each entry. Missing values were already filled in the "NaN", however any entries with missing or ‘NaN’ were dropped in all three datasets. The reason for that, as seen in Table 3, is the percentages of the features with missing values were very small or equal to 0%. All the data seemed to be consistent and accurate with the expectations of the datasets.

Figure 1
Grouped Histogram of Primary_Impact Type vs Season from Video_Review Dataset



As seen in Table 1, all three datasets contain a variety of variables such as numerical features, categorical features, and character strings. Standardization, using ‘StandardScaler’, was applied to all numerical features in each dataset to ensure all values were evaluated on the same

scale; for example, the value of 'total snaps' in the "Concussion" dataset may range up to 100, while the "Week of Injury" will be a maximum of 18. The model may be biased towards a large number so standardization ensures the models are equally evaluated. One-Hot Encoding was applied to the categorical variables in the datasets, such as player position and primary impact type, to transform these variables into numerical representations. This allowed for certain models, like clustering and classification, to be utilized and accurately evaluate the datasets. Due to the small size of the datasets, no additional features were applied to the datasets and no synthetic data was added as it could introduce noise that would interfere with model accuracy.

After pre-processing the datasets, the next step was to perform a univariate analysis. To begin, histograms and histogram variants for each feature of the dataset were created. For instance, in the "Video_Review" dataset, most of the histograms follow a left or right skewed distribution or are a bimodal distribution, not following a normal distribution. It was determined a player is more likely to get a concussion when they are tackling or being blocked. The impact typically is in the form of Helmet-to-Helmet or Helmet-to-Body. The results of helmet-to-helmet are not that surprising as the NFL has made a large effort to minimize plays that result in that type of hit, however it still is present in the game today. To further the NFL's understanding of the trends seen in these analyses so far, a grouped histogram was utilized to see how the different values compared to one another. Although the data only accounts for 2 years of NFL football games, understanding the trend surrounding how concussions occur is necessary to minimize the risk of it. In 2016, helmet-to-body was more frequently the cause of a concussion, however in 2017, helmet-to-helmet was the leading cause (Figure 1).

In addition, in 2017, there was a large jump in the frequency of tackling being the

cause of concussions in punt players, where in 2016 tackling/blocked had the same number of instances. When looking particularly at the relationship between the primary impact and player activity, it becomes clear for the areas that need to be addressed. When tackling, helmet-to -body is by far the most frequent cause of a concussion. When being blocked, helmet-to-helmet is the most frequent impact. Understanding these relationships and how the data changes season by season will allow the NFL to implement safety regulations to minimize concussions and can possibly use the data to fund research on different tackling/blocking techniques to limit the adverse events that can occur playing in the NFL.

The last analysis completed on the datasets was bivariate. For this analysis, correlation heat maps, scatter plots, pair plots, and line graphs were produced to highlight the data changing over time. During this analysis, heatmaps indicated that many variables had a high correlation with each other, especially in the “InjuryRecord” and “Concussion” datasets. This indicated that multicollinearity was present. This observation guided modeling decisions in future analysis as PCR and PLSR can combat multicollinearity using dimensionality reduction.

In this analysis, the “Video_Review” dataset was determined to be "usable" based on the strong correlation of PlayID and GSISID (-0.53). According to Kaggle, the PlayID is a "Numeric play identifier" and GSISID is "Unique player identification". During punt players, usually the personnel on each team is the same (the special team’s unit) so it would make sense there is a stronger correlation between these two columns. It will not be 100% correlation as players may vary depending if their team is punting or receiving (or if there are injuries at different parts of the game that impact who is on the field).

The outliers in this dataset are not that common however, there is one instance where the data lists the primary impact type as "unclear" and there are only two instances of "helmet to

ground" listed as the primary impact type. These are not the norm, as helmet-to-helmet or helmet-to-body are most frequent in the dataset. When looking at the correlation heat map, the highest correlations come between ‘Gamekey’ and ‘Season_Year’ (0.83), which does indicate some redundancy as GameKey is a "Numeric game identifier, unique across seasons"; in other words, the Gamekey is going to be in line with the season year. The other highest correlation came from the player_activity_dervied and the primary_partner_activity_derived; this is not necessarily the most insightful information and is expected since two players are not both going to be tackling each other - one will be blocking and the other will be tackling. When completing the line plots, the biggest trend that sticks out is that as time went on, the concussions related to helmet-to-helmet contact are increasing. Based on the NFL's recent initiatives to limit helmet-to-helmet contact by changing kick-offs, increasing penalty yardage and fines, I am not surprised the data resulted in this trend.

None of the datasets were unqualified due to issues in the data. All the data was cleaned and processed and had minimal duplicate values and all null values were filled in or dropped. In addition, all three datasets were standardized and encoded so all entries were equal in impact and numerical, making analysis such as mode, mean, median, easier for the model.

Section 5: Modeling/Analysis

Numerous predictive learning models were utilized such as classification, regression, feature engineering, and clustering to identify linear, non-linear, and complex patterns in the datasets. The use of a variety of modeling techniques allowed for diverse insights into how each dataset performed under certain conditions. The datasets all varied in size and the types of features they each consisted of so testing multiple models assisted in ensuring that the most accurate model was being used.

Each dataset was subject to data splitting. All three datasets had an 80/20 split applied to them in order to represent the training and testing sets. This split allowed for insight into how the models were generalizing to unseen data and help gain more insight into the model's performance. The EDA process outlined in Section 4 were applied to the training data then the test set; this helped to prevent overfitting.

Hyperparameter tuning was performed in all models used throughout this project. Hyperparameter tuning was used to help combat some of the flaws of the datasets, such as their size. Tuning allows for the models to be tested in an optimal condition and can provide a more robust analysis into how the model is performing. In regression models, such as PLSR and PCR, the number of components was that parameter being manipulated. This tuning helped control the intensity of dimensionality reduction; ideally, the number of components reached an ideal tradeoff between underfitting (not enough components to learn the relationships) and overfitting the model to noise (too many unnecessary features being used). In classification models, such as the decision tree and random forest models, hyperparameters were used to control how deep and wide the trees went. This was controlled using tools like 'max_depth', 'min_samples_leaf', and 'min_samples_split'. Again, an ideal bias-varience tradeoff needed to be made to ensure the model did not underfit the data but also did not overfit it. For clustering models, such as DBSCAN, tuning involves the neighborhood radius and minimum number of samples for a cluster. These parameters were 'eps' and 'min_samples' respectively and were selected using k-distance plots. Lastly, for HAC, tuning involved selecting the best linkage criteria to verify the clusters were combined in the most efficient and effective way.

For supervised machine learning models, a five fold cross-validation technique was applied. Because all three datasets were small, this metric was utilized to ensure different

subsets of the dataset were being reached and the model was not training on one specific pattern, helping to limit the risk of overfitting. The average of these five folds were used as the final performance metric. For unsupervised machine learning models, the primary metric used to evaluate the models' performance was silhouette score and accuracy. Tables 5 and 6 provide a summary of the appropriate evaluation metrics for all machine learning models used in this project.

The techniques with the most insightful results were from Partial Least Squares Regression (PLSR)/Principal Component Regression (PCR), decision trees/random forest regressions, and DBSCAN and Hierarchical Agglomerative Clustering (HAC). These models were selected due to the information learned in EDA and pre-processing. For example, when it comes to regression-based models, two of the datasets showed strong signs of multicollinearity. As a result, it was assumed PLSR and PCR would be a good model to train these datasets due to their ability to handle multicollinearity through dimensional reduction. Decision trees and random forest models provide insight into non-linear patterns or relationships in the datasets. In addition, they handle categorical features well, which was the data type of all three of the target features in this project ('Primary_Impact_Type', 'Surface', 'Reported_Injury_Type'). That made both of these techniques strong candidates for further analysis and model selection. Lastly, clustering models, like DBSCAN and HAC, allow for more contextual information regarding the datasets. In this project, that would be looking in particular at if there were common risk factors or features that may put a player at a higher risk for a concussion or lower limb injury, the goal of our project.

PLSR and PCR

When looking at PLSR, cross validation and standardization were applied since this model is sensitive to large outliers. Based on the RMSE scores, there are no obvious signs of

overfitting in the model. RMSE was used as it is a consistent and accurate way to determine the model's performance. For PLSR, n-components were used to limit the model's complexity (Scikit-learn, n.d.). It was interesting to see that forward selection in the "Concussion" dataset chose 11 features and had the highest RMSE out of the other models. Forward selection is typically a simpler model, especially compared to PCR and PLSR, so there might be some overfitting within that model.

Decision Trees and Random Forest Regressions

These two models are very prone to overfitting; therefore standardization and cross validation were used with both models to improve generalization. In addition, to help combat the complexity of the models, regulations such as cross validation was used to test how well the model performs on new unseen data; it is a way to see if overfitting is occurring before implementation a 'max_depth', 'min_samples_split', and 'min_samples_leaf' were introduced to prevent the code from continuing to run and overfitting to fit the data perfectly. The parameters were selected manually, however in the future, running a grid search could be beneficial and improve the performance of the decision tree and random forest. In addition, as described in Figure 1, a class imbalance does exist at least in the "Video_Review" dataset. To combat the class imbalance, a class weight of 'balanced' was applied during modeling. This process gives more 'weight' to minority classes in a dataset so the model can train without penalizing a feature that is not as common. The metrics used in these models to determine performance were mean CV accuracy, standard deviation, and training accuracy since these are classification models.

DBSCAN and Hierarchical Agglomerative Clustering (HAC)

In the DBSCAN, the use of a manually grid search for eps parameters using a k-distance plot was completed. This allowed for a clear visualization of what eps ranges were the best to test in the model. The model only utilized ranges that were slightly below or slightly above the

‘best fit’ point. This ensured that the model picked an eps parameter that balanced the bias and variance tradeoff. In addition, the use of noise ratio monitoring allowed for easy identification of signs of overfitting (as seen with the “Concussion” dataset). For HAC, the use of linkage comparison allowed for the selection of structure that would generalize well to unseen data. The metric used in both models are silhouette scores as it allows for a clear comparison to the k-means models and offers insight into how well the model clusters are fitting the data.

Section 6: Results

The techniques with the most insightful results were from Partial Least Squares Regression (PLSR)/Principal Component Regression (PCR), decision trees/random forest regressions, and DBSCAN and Hierarchical Agglomerative Clustering (HAC). Tables 5 and 6 provide a summary of the appropriate evaluation metrics for all machine learning models used in this project.

PLSR and PCR

All three datasets performed the best with PLSR. PLSR produced an CV RMSE average of 0.8241 for the “Video_Review” dataset, 0.4926 for the “InjuryRecord” dataset, and 0.3498 for the “Concussion” dataset - the lowest produced CV RMSEs’ out of all techniques for all three datasets; the average CV RMSE for this technique across all three datasets is 0.5555. PLSR performed well in these datasets, specifically “InjuryRecord” and “Concussion”, most likely due to the high multicollinearity present in the dataset which PLSR can combat with dimension reduction. As mentioned, the datasets are all small (the biggest is the “Concussion” dataset with 390 entries) which usually makes them prone to overfit. Forward and backward selection can lead to overfitting especially when there are numerous weak features (or not a lot of features in general) not strongly predicting the target value. This could explain why these stepwise methods performed the worst in all three datasets. PCR performed slightly worse than PLSR, with an

average CV RMSE of 0.5899 across all three datasets, but it still was successful in all three datasets (second best performer) but because PSLR can take PCR a set further by finding the best predictor in weak features, it wins out for all three datasets. As stated in the documentation named *Principal Component Regression vs Partial Least Squares Regression* on Scikit-learn, "PLS can outperform PCR when the target is strongly correlated with some directions in the data that have a low variance", which meets the expectations of this project precisely.

Decision Trees and Random Forest Regressions

The 'Concussion' dataset performed well with decision trees and random forest regressions; the decision tree produced at 90.62% cross validation accuracy with a standard deviation of 4.9% and the random forest regression produced an 84% cross validation accuracy and 4.7% standard deviation. The most important features in predicting concussions in these datasets were "Team", "Season", and "Average Playtime Before Injury" which had an importance level of about 0.27, 0.24, and 0.13 respectively. The higher the importance number, the more predictive the feature is. The next four features were "Games Missed", "Total Snaps", "Week of Injury", and "Opposing Team" which all had an importance value of about 0.07. This information indicates that team identity and the NFL season were predictive patterns in this dataset. This could suggest that there are differences in team's training procedures, coaching schemes and techniques or that rule changes and equipment redesign could be influencing the data. In addition, a player's playtime also has a strong influence on their risk for concussion. It is not surprising that the longer a player is on the field, that their likelihood of a concussion increases as there are more opportunities for the individual to be injured. However, comprehending when a player's playtime puts them into a high-risk situation, could allow the NFL to limit play time or consider shortening games. The 'Concussion' dataset is the largest of the three datasets with 390 entries. With a large dataset, these models are more likely to produce

well balanced classes with impactful features in relation to the target value (i.e. “Reported Injury Type”) which is why it is a strong stable model for classification.

The other two datasets did not perform as well. For the “Video_Review” dataset, the decision tree regression produced a 42.93% cross validation accuracy with a standard deviation of 16.7% and a cross-validation accuracy of 45.47% for random forest regression with a standard deviation of 17%. The most important features were “PlayID” and “Primary_Partner_GSISID” with both having an importance value of about 0.23. The “PlayID” feature is a numeric ID number that identifies the play the concussion occurred on; it would make sense that this feature would be highly influential in predicting the primary impact type that led to a concussion on punt plays. The “Primary_Partner_GSISID” is a unique player ID number that represents the impacting player involved in the helmet impact. Again, this directly correlates with the target variable of primary impact type so its importance value being higher is not surprising.

For the ‘InjuryRecord’, the decision tree produced a 42.40% cross validation accuracy with a 8.94% standard deviation and a 40.26% cross validation accuracy with a standard deviation of 9.18% for random forest regression. For this dataset, the top features were “PlayerKey” and “GameID” which both had an importance value of about 0.33. As explained above, ‘PlayerKey’ is a variable that is a unique five-digit number that identifies the NFL player who was injured. The “GameID” variable is a unique identifier of the game the player was playing in when injury occurred. Both features add information to the model about the game and player who suffered a lower limb injury so it is understandable they would be influential in predicting outcomes.

One major issue for both datasets is that the features should not have been label-encoded when training these models. When looking at the “Video_Review” dataset, it is shown that there is high variance and low accuracy. The model is most likely overfitting to noise due to lack of

features and entries, so it is not surprising it performs poorly. The “InjuryRecord” dataset had low accuracy but a moderate standard deviation. This indicates the model is not generalizing well; however, it is consistent. This reiterates that the connection between the features and target value are not very strong in this dataset.

DBSCAN and Hierarchical Agglomerative Clustering (HAC)

The first dataset, the “Video_Review”, produced a silhouette score of 0.2362 at k=4. Once again, the size of the dataset negatively impacts this model since it only created 4 clusters and produced a low silhouette score. The data loosely was able to form some separate clusters, but the data is not distinctly separated into them. In other words, the clusters are a bit ‘blurred’ together. In terms of our dataset, this indicates that there may be some distinct features that have a relationship with the target (primary impact type) but it is not a very strong one. For the second dataset, “InjuryRecord”, the silhouette score was 0.3616 at k=9. This score was an improvement compared to the first dataset and it also was able to form more clusters which indicates there is a stronger distinction between the features in this dataset. The silhouette score portrays that the clustering is moderately well defined but there may be some overlap between groups. This result is not that surprising considering that injuries in the NFL will have some overlapping data like position, injury type, etc. The “Concussion” dataset performed the worst with a silhouette score of 0.1436 at k=2. This silhouette score is close to zero which indicates that there are very weakly structured clusters in the dataset and many of the points will overlap. This dataset has consistently performed well in supervised learning models, which indicates that this data has more of a predictive relationship that should be explored further.

When performing DBSCAN on the “Video_Review” dataset, it was determined that an eps of 2.9 produced the best silhouette score of 0.17 with 2 clusters. This score is extremely low and highlights that the separation between the two groups is not very strong; the dataset structure

is weak and likely overlapping each other. When performing the HAC, it was determined that the ward linkage produced the best silhouette score of 0.21. This was a bit of an improvement over the DBSCAN but not by much. Although the size of the dataset is most likely influencing the results here, these metrics suggest that the concussions taking place on punts most likely do not fit into well-defined groupings, making addressing their frequency more difficult.

When looking at the "InjuryRecord" dataset, the DBSCAN did best with an eps of 2.0 using a minimum sample of ten and three clusters. As a result, it produced a silhouette score of 0.38, which is the best it has performed up to this point. However, the model did utilize 24 noise points indicating it did exclude about 25% of the dataset. This suggests that the dataset is not ideal for this type of model, and to make it perform better, the model did not include data that did not fit tightly into its clusters. When only using 1 noise point, the silhouette score drops to 0.28, a moderate drop off. Next, performing the HAC displayed that the ward linkage did the best with a 0.37 silhouette score. This suggests that the data can perform slightly distinct clusters, and in terms of this dataset, suggests there may be a relationship between specific injuries and playing surface.

Lastly, the "Concussion" dataset did not perform well with the DBSCAN. The best result came with an eps of 1.45 to produce a silhouette score of 0.49. Although that score is the best of all the datasets, it uses 310 noise points. In other words, it completely gets rid of most of the data points to perform better. As mentioned in the k-means section, this dataset does well with supervised learning and clusters, or distinct groups does not satisfy the model. The HAC produced an average silhouette score of 0.46 for single, complete, and average linkage and a 0.12 score for ward linkage. This indicates that the distance-based approaches are better suited for this

dataset most likely due to its poor ability to form distinct clusters. Table 4 provides a summary of all silhouette score results for K-Means, DBSCAN, and HAC.

***Table 4**: Summary Table of Silhouette Scores for K-Means, DBSCAN, and HAC*

Dataset	K-means	DBSCAN	HAC
Video_Review	0.2362 at k=4	0.17 with 2 clusters (eps=2.9)	0.21 (ward linkage)
Injury_Record	0.3616 at k=9	0.38 with 3 clusters (eps=2.0)	0.37 (ward linkage)
Concussion	0.1436 at k=2	0.49 with 3 clusters (eps=1.45)	0.46 (when using single, complete, and average linkage)

***Table 5**: Summary Table of CV RMSE for various machine learning models*

Dataset	Polynomial Terms	Interaction Terms	Lasso Regression	Ridge Regression	Elastic Net Regression	Forward Selection	Backward Selection	PCR	PLSR
Video_Review	2.595	2.636	1.07	1.25	1.07	1.2076	1.2076	0.9214	0.8241
Injury_Record	1.256	0.712	0.51	0.54	0.51	0.5076	0.5208	0.4972	0.4926
Concussion	0.384	0.403	0.40	0.37	0.36	0.3721	0.3708	0.3513	0.3498

***Table 6**: Summary Table of CV accuracy for various machine learning models*

Dataset	Logistic Regression	Support vector machine (with best C)	Decision tree regression	Random Forest Regression	KNN	Gradient Boosting
Video_Review	0.5946	0.7586	0.4293	0.4547	0.6250	0.5533
Injury_Record	0.5048	0.5357	0.4240	0.4026	0.4770	0.5346
Concussion	0.8031	0.9500	0.9062	0.8400	0.8460	0.8885

Section 7: Recommendations

The “Video_Review” and “InjuryRecord” datasets never really produced any strong results. The biggest problem with the datasets is their size. If more data was collected over a longer time, the conclusions could be more concrete and the recommendations for interventions would be more specific. However, based on the insights produced from the “Video_Review” dataset (the accuracy for this dataset was around 50-60% or had an average RMSE of 1.41), the indication is that features like friendly fire, player activity, and turnover related injuries have

limited predictive value regarding primary impact type. By being unable to predict what factors affect primary impact type, it is harder to recommend rule changes or promote different coaching techniques in preventing injury.

Similarly, after each analysis, it was often indicated that the features in the "InjuryRecord" dataset did not have a strong enough relationship to the target, so the model performed poorly (the highest accuracy was 65% across all models and lowest RMSE was 0.4926). This indicated that features like body parts injured, and length of time missed due to injury did not have a strong relationship with the surface the injury occurred on. As discussions regarding the dangers of synthetic turf continue in the NFL, this analysis disapproved the need to invest in natural grass and can save the league time and money on changing the field if there is no real prevention measure being achieved. Again, more data on the impact and severity of player's injuries needs to be collected for a definite recommendation to be made.

The "Concussion" dataset performed well in classification models. This indicates that the target 'Reported Injury Type' has a clear relationship with features like number of snaps into the game when concussion occurs, a player's position, and the average playtime before injury. Based on these insights, implementing protective equipment for head injuries, specifically for cornerbacks, wide receivers, and safeties who experienced the most injuries in the 2012-2014 NFL seasons is vital. This could focus on re-designing helmets, exploring different tackling techniques, or even considering dissolving contact in the game of football and moving towards similar rules to flag football. Another option would be to shorten the NFL season or shorten the length of games to minimize the number of repetitive contact players face. Additionally, the NFL could use the data regarding average playtime prior to injury (i.e. a concussion) to comprehend when a player's playtime starts to put them at a high risk for injury. This could influence the

NFL to put players on a playtime limit as it is not surprising that the longer a player is on the field, that their likelihood of a concussion increases as there are more opportunities for the individual to be injured.

As stated throughout this report, all three datasets were small. The “Video_Review” dataset had 37 entries, the “InjuryRecord” dataset had 105 entries, and the “Concussion” dataset was the biggest with 390 entries. Small datasets mean the machine learning models are more prone to overfitting and produce less accurate results. In addition, the most recent dataset used was published in 2019. Having a full dataset from more recent seasons, especially since the adaption of the dynamic kickoff, helmet re-design, and new penalties for contact to players head and lower bodies, would provide the project with a more accurate insight and in turn recommendations.

Moving forward, implementing risk studies, A/B testing of coaching interventions, and longitudinal study of injury trends in the NFL would be beneficial. Risk studies would help determine and measure what actions are the riskiest in NFL games. A/B testing of coaching interventions would allow a direct comparison between two different approaches to see which is best in preventing injuries. Longitudinal studies would provide more information and accurate insight into how the NFL’s injury rates have changed over multiple years and identify areas of concern. An opportunity to also collect and analyze data regarding the NFL’s return-to-play policy, especially after a concussion, would bring more insight into the health and safety of players in the league. Understanding how the NFL clears their players to return to play and how these decisions affect the likelihood of repeated concussions, or the development of Chronic Traumatic Encephalopathy (CTE) would not only improve players’ wellbeing but could prevent the NFL from facing additional lawsuits from current and former players.

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Appendix
Jupyter Notebook Links

Jupyter Notebook Topic	**Link to Notebook**
Preprocessing	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week2.ipynb
Univariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week4.ipynb
Bivariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week6.ipynb
Multivariate Analysis	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week9.ipynb
Basic Modeling	https://github.com/hnm15/ModB/blob/cd4113c89dad2d7afab037c0bf91dddd932673b7/Week11.ipynb
Linear Regression Part 1	https://github.com/hnm15/capstone/blob/8d6cc89716a283cc60a4ed782a07c702c3cf354f/Wk1-JupyterNotebook-Metcalf.ipynb
Linear Regression Part 2	https://github.com/hnm15/capstone/blob/fe79c2d4a529a3097d9db10da310041c0c4f90bf/Wk2-JupyterNotebook-Metcalf.docx.ipynb
Linear Regression Part 3	https://github.com/hnm15/capstone/blob/20d2b93736b1bb7b37b766d1b7e4cd1203f5a9f9/Wk3-JupyterNotebook-Metcalf.ipynb
Logistic Regression and Feature Scaling	https://github.com/hnm15/capstone/blob/2a5f2cc1ec6d00b7d950f4f1fcbdac93be7a1bf4/Wk4-JupyterNotebook-Metcalf.ipynb
Support Vector Machines	https://github.com/hnm15/capstone/blob/fe79c2d4a529a3097d9db10da310041c0c4f90bf/Wk5-JupyterNotebook-Metcalf.ipynb
Decision Trees and Random Forests	https://github.com/hnm15/capstone/blob/fe79

	c2d4a529a3097d9db10da310041c0c4f90bf/Wk6-JupyterNotebook-Metcalf.ipynb
K-Nearest Neighbors	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk8-JupyterNotebook-Metcalf.ipynb
Gradient boost, learning rate, number of estimators, tree depth, and regularization	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk9-JupyterNotebook-Metcalf.ipynb
K-means: Elbow method, Silhouette score, Feature scaling	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk10-JupyterNotebook-Metcalf.ipynb
DBSCAN, HAC, Linkage methods in HAC, Dendrograms	https://github.com/hnm15/capstone/blob/9e1ca4752205cc1251a6a945807a6c406e6da906/Wk11-JupyterNotebook-Metcalf.ipynb